```
C:\Users\Laptop\cloudsim-3.0.3\examples>java -version
java version "1.8.0_202"
Java(TM) SE Runtime Environment (build 1.8.0_202-b08)
Java HotSpot(TM) 64-Bit Server VM (build 25.202-b08, mixed mode)

C:\Users\Laptop\cloudsim-3.0.3\examples>javac -version
javac 1.8.0_202

C:\Users\Laptop\cloudsim-3.0.3\examples>ant -version
Apache Ant(TM) version 1.10.15 compiled on August 25 2024

C:\Users\Laptop\cloudsim-3.0.3\examples>
```

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Laptop>cd C:\Users\Laptop\cloudsim-3.0.3

C:\Users\Laptop\cloudsim-3.0.3>ant
Buildfile: C:\Users\Laptop\cloudsim-3.0.3\build.xml

prepare:

build:
    [javac] Compiling 134 source files to C:\Users\Laptop\cloudsim-3.0.3\classes

makejar:
     [echo] Compiling a new jar file, named: "cloudsim-new.jar".
     [echo] This jar file contains CloudSim classes only.
      [jar] Building jar: C:\Users\Laptop\cloudsim-3.0.3\jars\cloudsim-new.jar

BUILD SUCCESSFUL
Total time: 7 seconds

C:\Users\Laptop\cloudsim-3.0.3>cd examples

C:\Users\Laptop\cloudsim-3.0.3\examples>javac -cp "..\jars\cloudsim-new.jar;..\jars\commons-math3-3.2.jar" org\cloudbus\cloudsim\examples\CloudSimExample1.java

C:\Users\Laptop\cloudsim-3.0.3\examples>java -cp ".;..\jars\cloudsim-new.jar;..\jars\commons-math3-3.2.jar" org.cloudbus.cloudsim.examples.CloudSimExample1
Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 resource(s)
0.0: Broker: Trying to Create VM #0 in Datacenter_0
0.1: Broker: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker: Sending cloudlet 0 to VM #0
400.1: Broker: Cloudlet 0 received
400.1: Broker: All Cloudlets executed. Finishing...
400.1: Broker: Destroying VM #0
Broker is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_0 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.

========== OUTPUT ==========
Cloudlet ID    STATUS    Data center ID    VM ID    Time    Start Time    Finish Time
     0         SUCCESS       2              0        400       0.1         400.1
CloudSimExample1 finished!

C:\Users\Laptop\cloudsim-3.0.3\examples>
```

```
C:\Users\Laptop\cloudsim-3.0.3\examples>javac -cp "..\jars\cloudsim-new.jar;..
\jars\commons-math3-3.2.jar" org\cloudbus\cloudsim\examples\TwoDatacenterCloud
Sim.java

C:\Users\Laptop\cloudsim-3.0.3\examples>java -cp ".;..\jars\cloudsim-new.jar;.
.\jars\commons-math3-3.2.jar" org.cloudbus.cloudsim.examples.TwoDatacenterClou
dSim
Initialising...
Starting CloudSim version 3.0
Datacenter_1 is starting...
Datacenter_2 is starting...
Broker_1 is starting...
Entities started.
0.0: Broker_1: Cloud Resource List received with 2 resource(s)
0.0: Broker_1: Trying to Create VM #0 in Datacenter_1
0.0: Broker_1: Trying to Create VM #1 in Datacenter_1
0.1: Broker_1: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker_1: VM #1 has been created in Datacenter #2, Host #0
0.1: Broker_1: Sending cloudlet 0 to VM #0
0.1: Broker_1: Sending cloudlet 1 to VM #1
10.1: Broker_1: Cloudlet 0 received
10.1: Broker_1: Cloudlet 1 received
10.1: Broker_1: All Cloudlets executed. Finishing...
10.1: Broker_1: Destroying VM #0
10.1: Broker_1: Destroying VM #1
Broker_1 is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_1 is shutting down...
Datacenter_2 is shutting down...
Broker_1 is shutting down...
Simulation completed.
Simulation completed.

========== Cloudlet Execution Results ==========
Cloudlet ID | Status | VM ID | Start Time | Finish Time
0           | SUCCESS | 0     | 0.10       | 10.10
1           | SUCCESS | 1     | 0.10       | 10.10

C:\Users\Laptop\cloudsim-3.0.3\examples>
```

```
C:\Users\Laptop\cloudsim-3.0.3\examples>javac -cp "..\jars\cloudsim-new.jar;
..\jars\commons-math3-3.2.jar" org\cloudbus\cloudsim\examples\SimpleCloudSim
.java

C:\Users\Laptop\cloudsim-3.0.3\examples>java -cp ".;..\jars\cloudsim-new.jar
;..\jars\commons-math3-3.2.jar" org.cloudbus.cloudsim.examples.SimpleCloudSi
m
Initialising...
Starting CloudSim version 3.0
Datacenter_1 is starting...
Broker_1 is starting...
Entities started.
0.0: Broker_1: Cloud Resource List received with 1 resource(s)
0.0: Broker_1: Trying to Create VM #0 in Datacenter_1
0.1: Broker_1: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker_1: Sending cloudlet 0 to VM #0
0.1: Broker_1: Sending cloudlet 1 to VM #0
10.1: Broker_1: Cloudlet 0 received
10.1: Broker_1: Cloudlet 1 received
10.1: Broker_1: All Cloudlets executed. Finishing...
10.1: Broker_1: Destroying VM #0
Broker_1 is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_1 is shutting down...
Broker_1 is shutting down...
Simulation completed.
Simulation completed.

========== Cloudlet Execution Results ==========
Cloudlet ID | Status | VM ID | Start Time | Finish Time
0           | SUCCESS | 0     | 0.10       | 10.10
1           | SUCCESS | 0     | 0.10       | 10.10

C:\Users\Laptop\cloudsim-3.0.3\examples>
```

```
C:\Users\Laptop\cloudsim-3.0.3\examples>javac -cp "..\jars\cloudsim-new.jar;..
\jars\commons-math3-3.2.jar" org\cloudbus\cloudsim\examples\MultiVMCloudSim.ja
va

C:\Users\Laptop\cloudsim-3.0.3\examples>java -cp ".;..\jars\cloudsim-new.jar;.
.\jars\commons-math3-3.2.jar" org.cloudbus.cloudsim.examples.MultiVMCloudSim
Initialising...
Starting CloudSim version 3.0
Datacenter_1 is starting...
Broker_1 is starting...
Entities started.
0.0: Broker_1: Cloud Resource List received with 1 resource(s)
0.0: Broker_1: Trying to Create VM #0 in Datacenter_1
0.0: Broker_1: Trying to Create VM #1 in Datacenter_1
0.1: Broker_1: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker_1: VM #1 has been created in Datacenter #2, Host #0
0.1: Broker_1: Sending cloudlet 0 to VM #0
0.1: Broker_1: Sending cloudlet 1 to VM #1
10.1: Broker_1: Cloudlet 0 received
10.1: Broker_1: Cloudlet 1 received
10.1: Broker_1: All Cloudlets executed. Finishing...
10.1: Broker_1: Destroying VM #0
10.1: Broker_1: Destroying VM #1
Broker_1 is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_1 is shutting down...
Broker_1 is shutting down...
Simulation completed.
Simulation completed.

========== Cloudlet Execution Results ==========
Cloudlet ID | Status | VM ID | Start Time | Finish Time | Execution Time
0           | SUCCESS | 0     | 0.10       | 10.10       | 10.00
1           | SUCCESS | 1     | 0.10       | 10.10       | 10.00

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```